

SCHOOL OF COMPUTER SCIENCE AND IT

YBN UNIVERSITY RANCHI

JAVA PROGRAMMING

ASSIGNMENT 03

1) Print Numbers from 0 to 10 and 10 to 0 Using Two While Loops

Description: Write a Java program that prints numbers from 0 to 10 using one `while` loop and then prints numbers from 10 to 0 using another `while` loop.

Hint:

- Use the first `while` loop to increment a counter from 0 to 10 and print each value.
- Use the second `while` loop to decrement the counter from 10 to 0 and print each value.

2) Sum of Positive Integers Until User Enters 0

Description: Write a Java program that prompts the user to input a series of integers until they enter 0. Calculate and print the sum of all the positive integers entered using a `while` loop.

Hint:

- Use a `while` loop to repeatedly prompt the user for input.
- Add only the positive integers to a sum variable.
- Terminate the loop when the user inputs 0.

3) Calculate Product of Numbers from 1 to 5

Description: Write a Java program that calculates the product of numbers from 1 to 5 using a `while` loop.

Hint:

- Initialize a product variable to 1.
- Use a `while` loop to multiply the product variable by each number from 1 to 5.

4) Detect Duplicate Input

Description: Write a Java program that prompts the user to input a series of numbers until they input a duplicate number. Use a `while` loop to check for duplicates.

Hint:

- Use a `HashSet` to store unique numbers.
- In each iteration of the `while` loop, check if the number is already in the set.

- If it is a duplicate, terminate the loop; otherwise, add it to the set.

5) Calculate Factorial of a Number

Description: Write a Java program that prompts the user to enter a positive integer and calculates its factorial using a `while` loop.

Hint:

- Initialize a factorial variable to 1.
- Use a `while` loop to multiply the factorial variable by each number down to 1.

6) Sum of Cubes of Even Numbers Up to a Specified Limit

Description: Write a Java program that calculates and prints the sum of the cubes of even numbers up to a specified limit (e.g., 20) using a `while` loop.

Hint:

- Use a `while` loop to iterate through even numbers up to the limit.
- Calculate the cube of each even number and add it to a sum variable.

7) Print First 10 Fibonacci Numbers

Description: Write a Java program that finds and prints the first 10 Fibonacci numbers using a `while` loop.

Hint:

- Initialize the first two Fibonacci numbers as 0 and 1.
- Use a `while` loop to calculate and print the next Fibonacci number by adding the two previous numbers.
- Continue this until you have printed 10 numbers.